

- 1 (a) (i) angle subtended at centre of circle B1
 arc equal in length to the radius B1 [2]
- (ii) arc = $r\theta$ and for one revolution, arc = $2\pi r$ M1
 so, $\theta = 2\pi r/r = 2\pi$ A0 [1]
- (b) (i) *either* weight provides/equals the centripetal force B1
or acceleration of free fall is centripetal acceleration B1
 $9.8 = 0.13 \times \omega^2$ M1
 $\omega = 8.7 \text{ rad s}^{-1}$ A0 [2]
- (ii) force in cord = weight + centripetal force (*can be an equation*) C1
 force in cord = $(L - 13) \times 5/1.8$ *or* force constant = $5.0/1.8$ C1
 $(L - 13) \times 5/1.8 = 5.0 + 5/9.8 \times L \times 10^{-2} \times 8.7^2$ C1
 $L = 17.2 \text{ cm}$ A1 [4]
 (*constant centripetal force of 5.0 N gives $L = 16.6 \text{ cm}$ allow 2/4*)

2 (a) (i) $pV = nRT$